THUSHARA SANDEEPANI

Computer Science Undergraduate

0711964546 thusharasvidanage@gmail.com Rathnapura, Sri Lanka

GitHub LinkedIn Portfolio

Profile

Computer Science undergraduate with a strong interest in software engineering, artificial intelligence, web development, and emerging technologies. Possesses foundational knowledge of programming, frontend and backend development, databases, REST APIs, and software development practices. Familiar with technologies such as Python, Java, JavaScript, React.js, Node.js, SQL, Git, and GitHub. A motivated, adaptable, and fast learner with a strong willingness to explore new technologies, solve problems, and develop practical solutions. Seeking an IT internship to apply technical knowledge, gain hands-on industry experience, and contribute to innovative and reliable technology solutions across software development, AI, web development, and other IT-related fields.

Education

University of Vavuniya
BSc in Computer Science

University of Colombo
Bachelor of Information Technology

Technical Skills

Programming Languages	Java, Python, C++, JavaScript, SQL, C, Shell
Web Technologies	HTML, CSS, JavaScript, React.js, Node.js, Bootstrap
Databases	MySQL, PostgreSQL, SQLite
Development Tools	Git, GitHub, Visual Studio Code, npm
Core Areas	Object-Oriented Programming, Database Management, Web Development, Software Development, Problem Solving

Projects

AeroEats – Smart Multi-Mode Delivery System *Personal Project*

- Developed a smart delivery platform concept supporting multiple delivery modes including drones, bikes, tuk-tuks, vans, and cars.
- Designed the system to support food, grocery, and parcel delivery services.
- Designed a web-based architecture with separate frontend and backend components.
- Worked with JavaScript, Node.js, databases, REST APIs, and modern web development technologies.
- Designed multiple payment options including Cash on Delivery and bank transfer.

Personal Portfolio Website *Personal Project*

- Designed and developed a responsive personal portfolio website to showcase education, technical skills, projects, and career interests.
- Used HTML, CSS, JavaScript, and responsive web design techniques.
- Integrated GitHub and LinkedIn profiles to create a professional online presence.
- Deployed the portfolio website using GitHub Pages.

Expense Tracker *Group Project*

Authentication Frontend Developer | React.js, Axios, React Router, JWT, Git/GitHub

- Developed responsive **Login and Registration pages** using React.js with reusable authentication components.
- Implemented frontend authentication flow including **form validation, JWT token handling, protected routes, and localStorage-based session management**.
- Integrated frontend authentication with REST APIs using **Axios**, handling login, registration, logout, loading states, and API errors.
- Collaborated with team members using **Git/GitHub feature branches and pull requests** to integrate frontend changes with the project.

“‘latex

Academic Research Project

Dengue Virus Serotype and Strain Variation Classification Using Alignment-Free K-Mer Feature Extraction and Machine Learning Classifiers

BSc Honours in Computer Science, University of Vavuniya 2026 – Present

- Developing a machine learning approach to classify Dengue virus serotypes (DENV-1 to DENV-4) using viral envelope protein sequence data.
- Processing FASTA sequence data and metadata for machine learning analysis, including quality control, duplicate removal, and handling incomplete or ambiguous sequences.
- Extracting alignment-free k-mer features (2-mer and 3-mer) from amino-acid sequences using Python and BioPython.
- Exploring physicochemical features including hydrophobicity, net charge, and molecular weight to improve classification performance.
- Evaluating Random Forest, SVM, XGBoost, and 1D-CNN models using accuracy, precision, recall, and F1-score.
- Investigating temporal generalisation by training models on historical sequences and evaluating their ability to classify more recent viral variants.
- **Technologies:** Python, BioPython, Pandas, NumPy, Scikit-learn, and PyTorch.

“‘

Relevant Coursework

- Object-Oriented Programming
- Software Engineering
- Data Structures and Algorithms
- Human-Computer Interaction
- Database Management Systems
- Web Development
- Computer Networks
- Statistics and Data Analysis

Certifications

Introduction to Kubernetes
in simpli learn skillup

Prompt Engineering Applications
in simpli learn skillup

An Introduction to JIRA Architecture
in Alison (Advance Learning Interactive Systems Online)

Soft Skills

- Problem Solving
- Analytical Thinking
- Self Learning
- Teamwork
- Time Management
- Attention to Detail
- Communication
- Adaptability
- Leadership

Languages

Sinhala Native
English Professional Working Proficiency

References

Dr. Nagulan
Senior Lecturer
Department of Physical Science
University of Vavuniya
+94 77 844 3891
nagulanr@vau.ac.lk

Lasantha Gunarathna
Senior software engineer
Xelleron pvt ltd
+94 76 255 0909